



Data Warehousing and Business Intelligence
(IT3021)
Assignment 2

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Step 1: Data Source for Assignment 2

The data warehouse implemented and loaded with data in Assignment 1 is used as the data source for Assignment 2. The data warehouse is built on the Parking Management dataset sourced from Kaggle, consisting of approximately one million parking transaction records spanning from several years.

1.1 Data Warehouse Overview

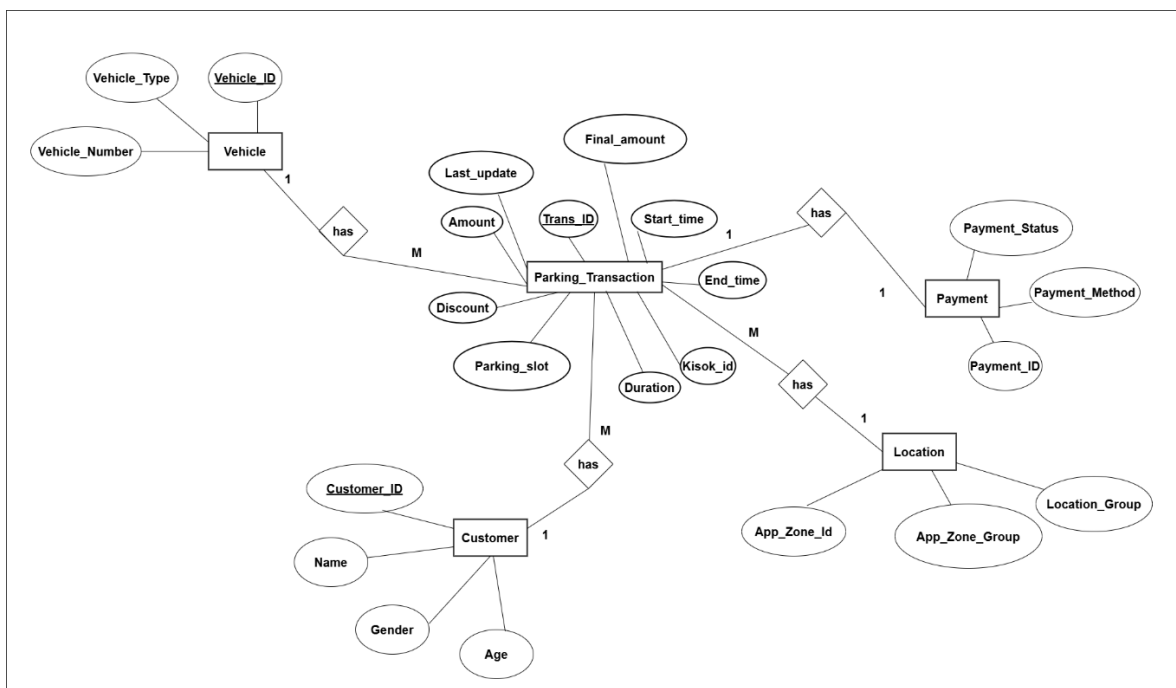
The data warehouse (Parking_DW) is designed using a Star Schema with a central fact table (FactParking) connected to five dimension tables. This structure enables fast and efficient OLAP querying and analysis.

1.2 Schema Description

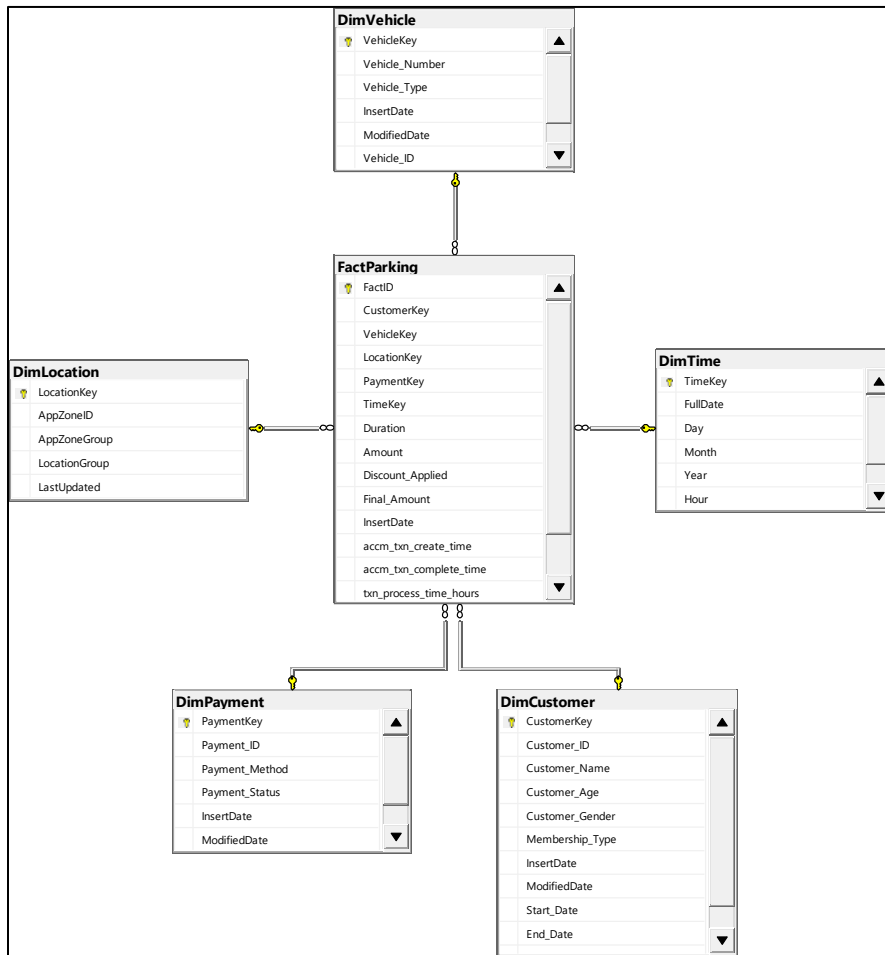
The Parking_DW data warehouse uses a star schema comprising the following tables:

- FactParking – Central fact table storing transactional measures: Amount, Discount_Applied, Final_Amount, Duration, and the accumulating timestamps.
- DimCustomer – Customer dimension with SCD Type 2 implementation to track membership changes over time.
- DimVehicle – Vehicle dimension containing vehicle type and number information.
- DimLocation – Location dimension with a three-level hierarchy: Location Group → App Zone Group → App Zone.
- DimPayment – Payment dimension capturing payment method and status.
- DimTime – Time dimension with date hierarchy: Year → Month → Day → Hour.

1.3 ER Diagram



1.4 Data Warehouse Star Schema



1.5 Data Source Tables

The data warehouse was populated from five source files that were first loaded into a staging database (Parking_Staging) before ETL transformations were applied:

File Name	Format	Contents
Parking_Transaction.csv	CSV	Transaction ID, Source, Duration, Start/End Time, Amount, Discount, Final Amount, Customer ID, Kiosk ID, App Zone, Payment ID
Payments.csv	CSV	Payment ID, Payment Method, Payment Status
Locations.xlsx	Excel	App Zone ID, App Zone Group, Location Group (three-level hierarchy)
Vehicles.xlsx	Excel	Vehicle ID, Vehicle Type, Vehicle Number, Customer ID
Customers.xlsx	Excel	Customer ID, Customer Name, Gender, Age, Member Type

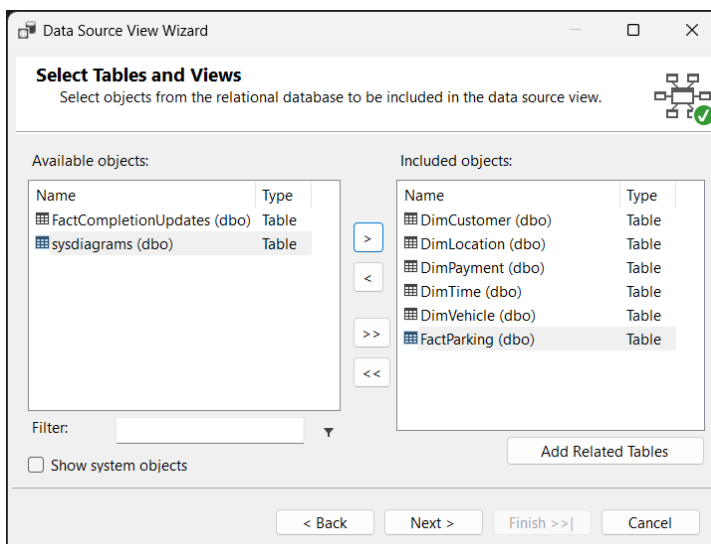
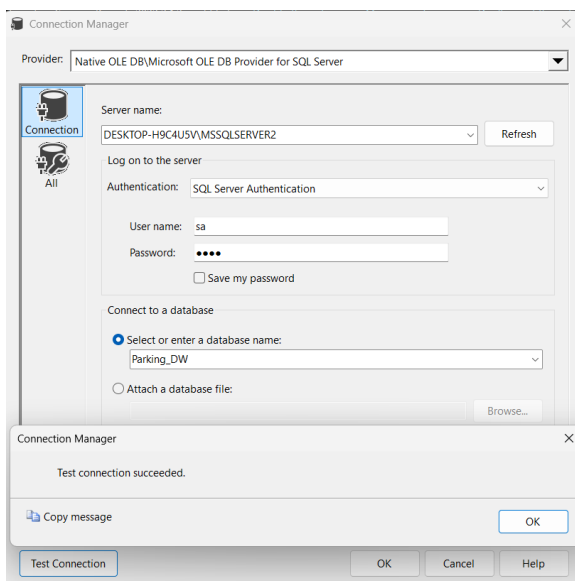
Step 2: SSAS Cube Implementation

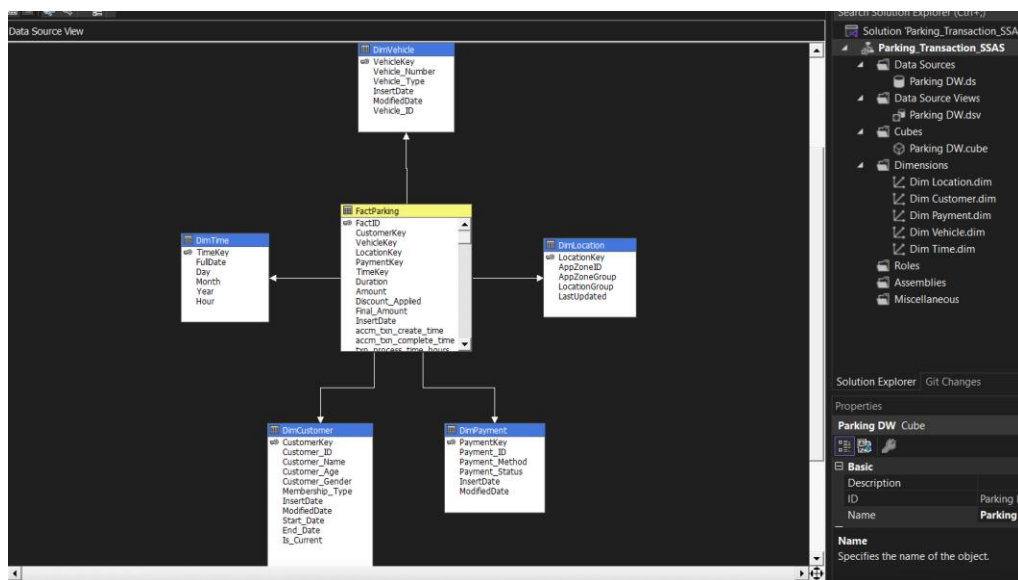
An SQL Server Analysis Services (SSAS) cube was created using the Parking_DW data warehouse as the data source. The cube enables multidimensional analysis of parking transaction data through measures and dimensions with hierarchies.

2.1 Creating the SSAS Project

The following steps were followed to create the SSAS cube project:

- Opened SQL Server Data Tools (SSDT) and created a new Analysis Services Multidimensional and Data Mining Project.
- Named the project and configured the target Analysis Services instance.
- Created a new Data Source pointing to the Parking_DW SQL Server database.
- Created a Data Source View (DSV) including all dimension tables and the FactParking fact table.
- Added named relationships between the fact table and dimension tables within the DSV.





2.2 Cube Structure – Measures

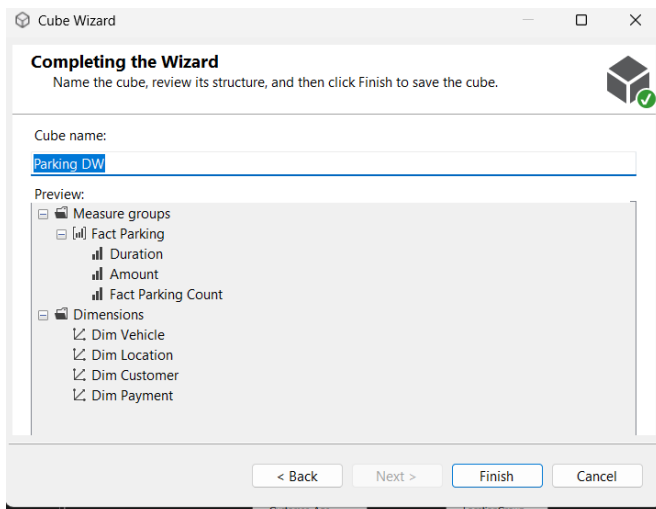
The following measures were included in the SSAS cube from the FactParking fact table:

- Amount – Total parking charge amount
- Discount_Applied – Total discount applied to the transaction
- Final_Amount – Final amount after discount
- Duration – Duration of the parking session (in minutes)
- Fact Parking Count – Count of transactions

2.3 Cube Structure – Dimensions

The following dimensions were connected to the cube:

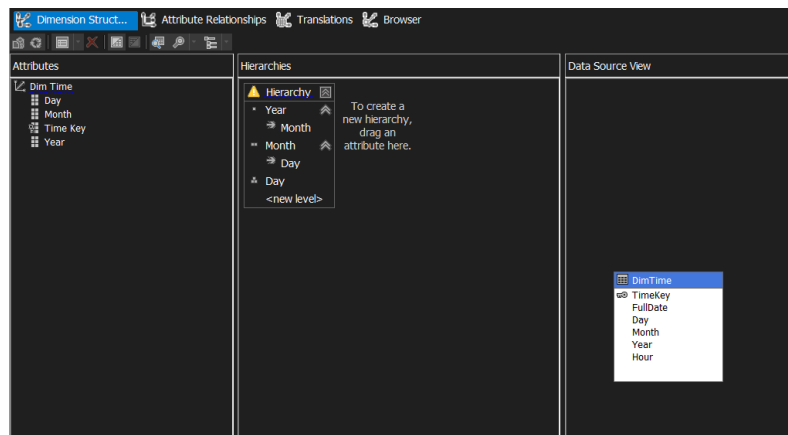
- Dim Customer – Customer details including membership type
- Dim Vehicle – Vehicle information (type, number)
- Dim Location – Location hierarchy (Location Group → App Zone Group → App Zone)
- Dim Payment – Payment method and status
- Dim Time – Date/time hierarchy (Year → Month → Day → Hour)



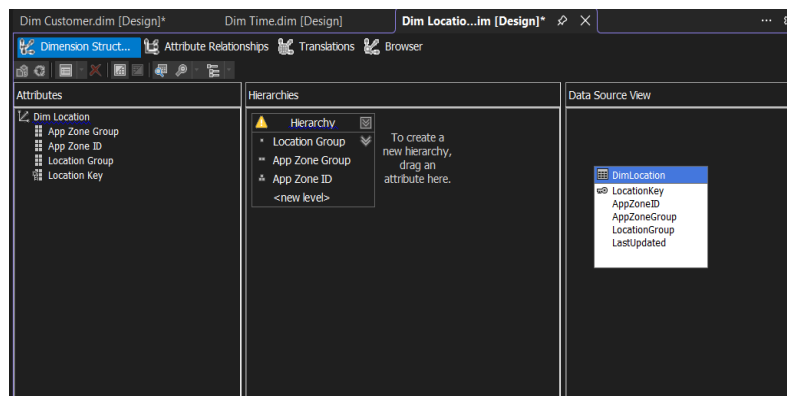
2.4 Hierarchy Implementation

The following hierarchies were implemented in the cube:

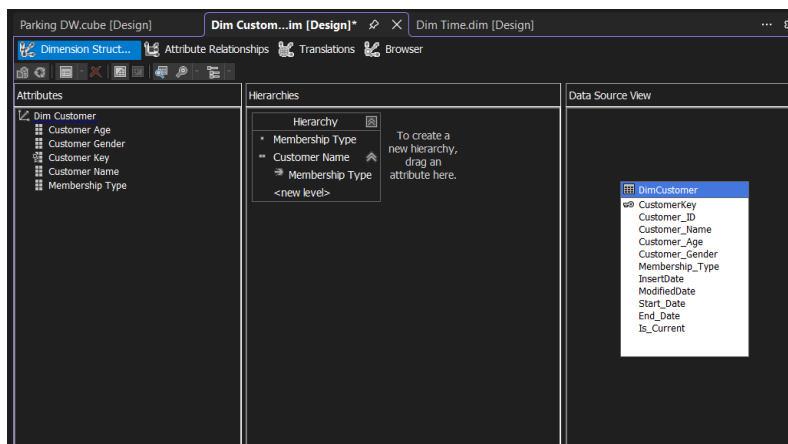
- Time Hierarchy (in Dim Time): Year → Month → Day → Hour – enables roll-up and drill-down analysis across time periods.



- Location Hierarchy (in Dim Location): Location Group → App Zone Group → App Zone – enables geographic drill-down analysis.

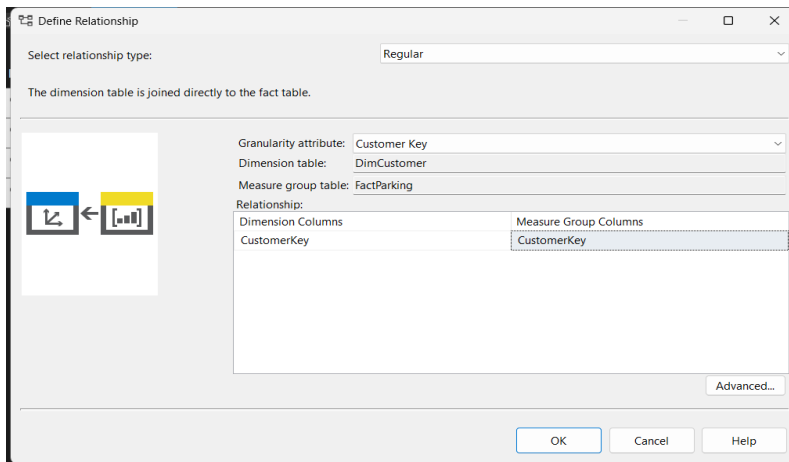


- Customer Hierarchy (in Dim Customer): Customer Name → Membership Type

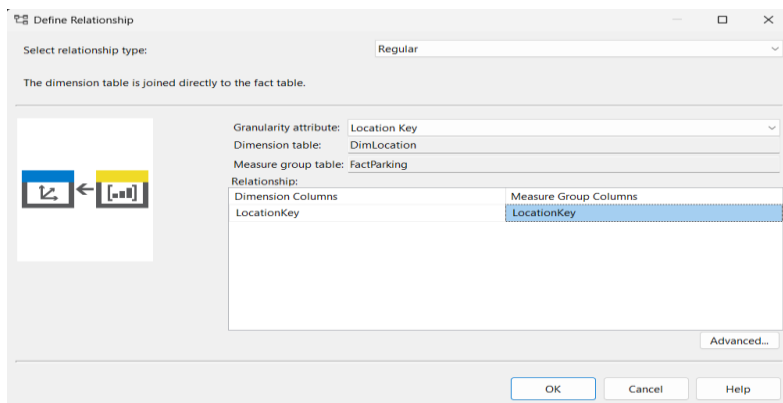


2.5 Relationship Demonstration

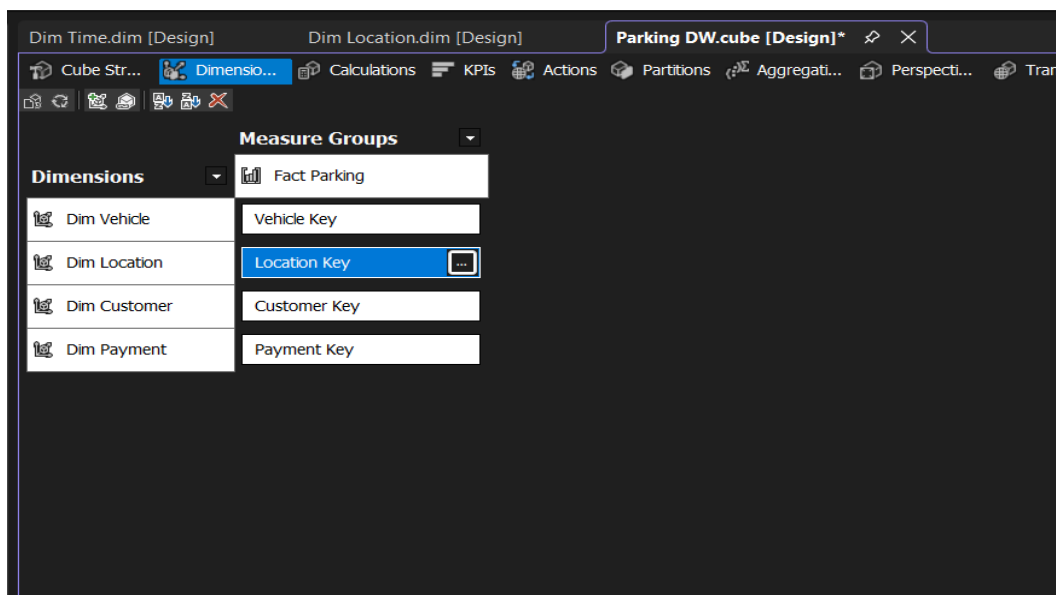
- Relationship between the Customer dim table and FactParking was created



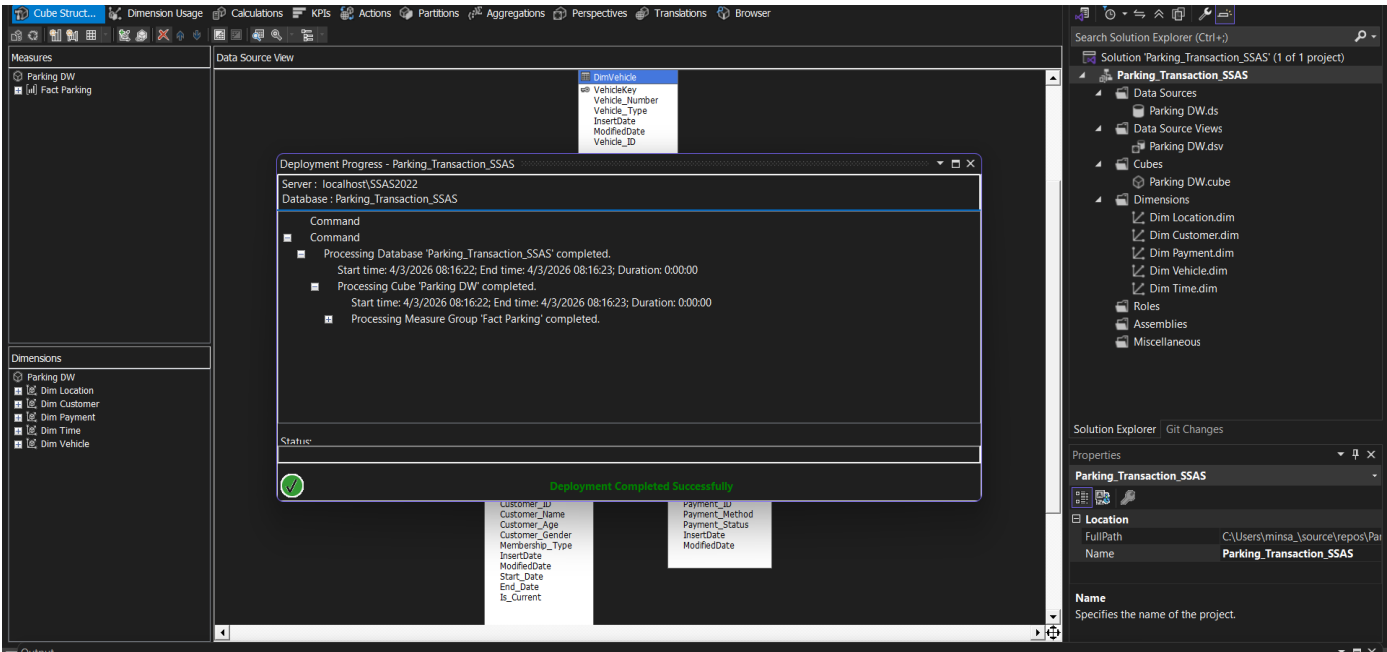
- Relationship between the Location dim table and FactParking was created.



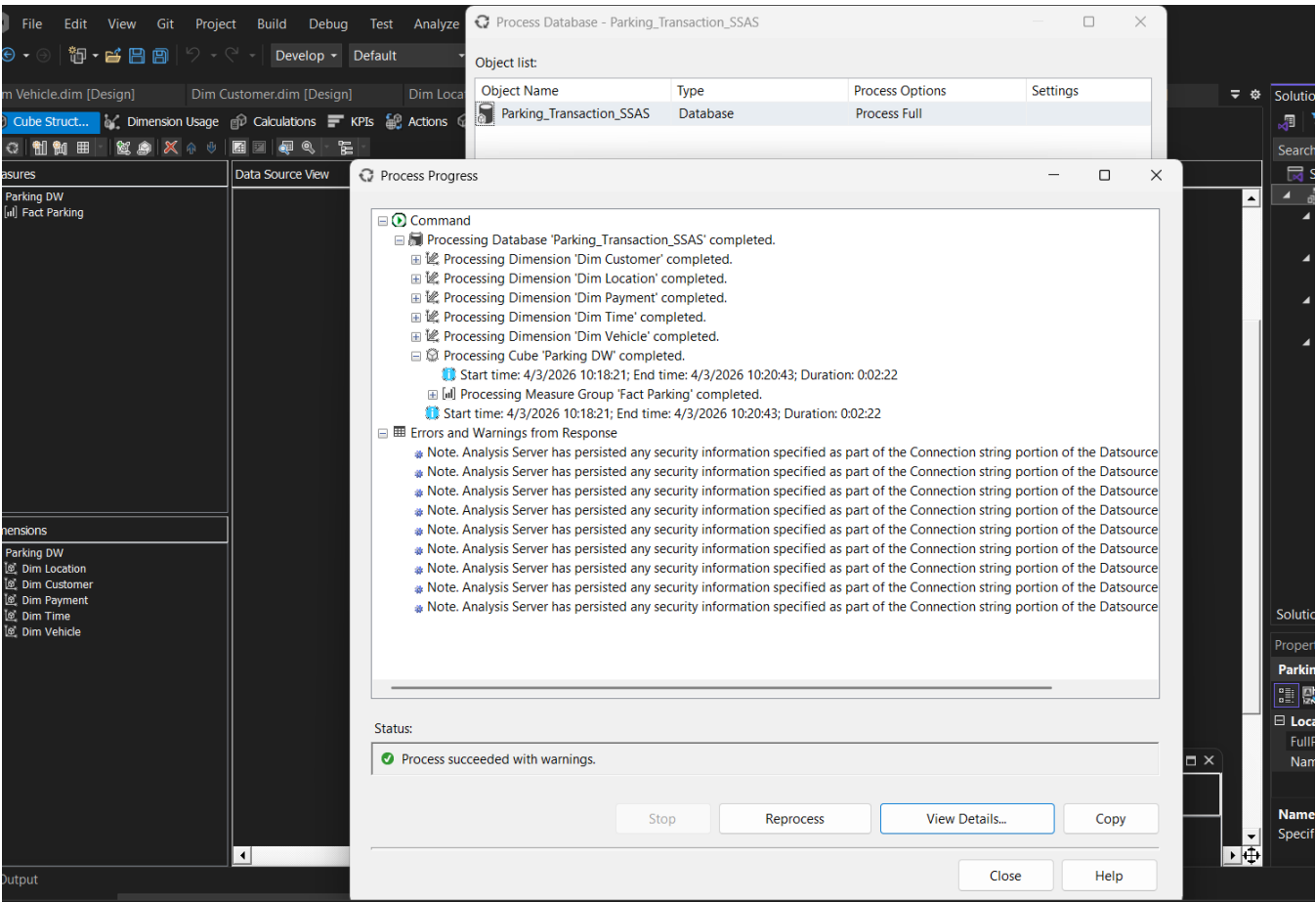
- Relationship between attributes were created



Then the cube was successfully deployed,



The project was processed successfully,



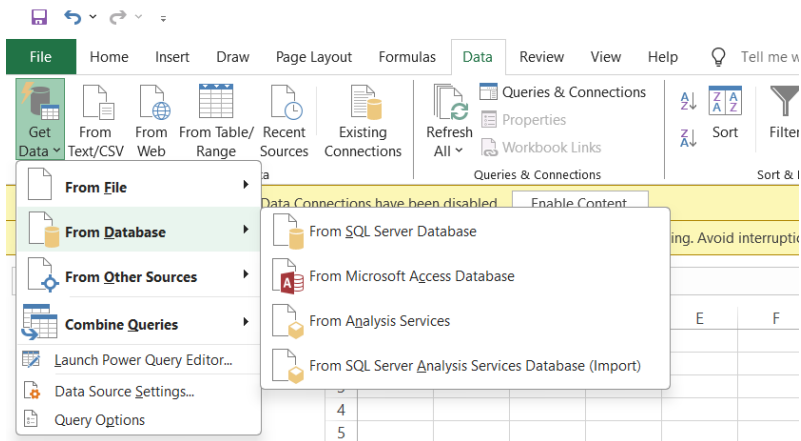
Step 3: Demonstration of OLAP Operations

An Excel workbook was connected to the deployed SSAS cube. Pivot Tables and Pivot Charts were used to demonstrate the five core OLAP operations using parking transaction data.

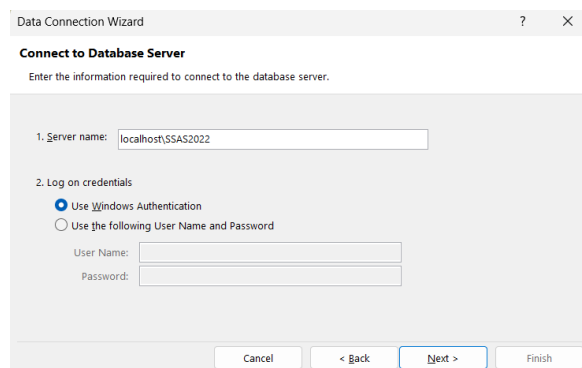
3.1 Connecting Excel to the SSAS Cube

The following steps were performed to connect Excel to the cube:

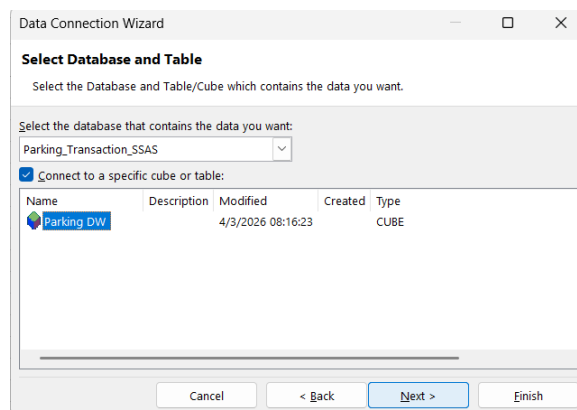
- Opened Excel and navigated to the Data tab.
- Selected "From Other Sources" → "From Analysis Services".



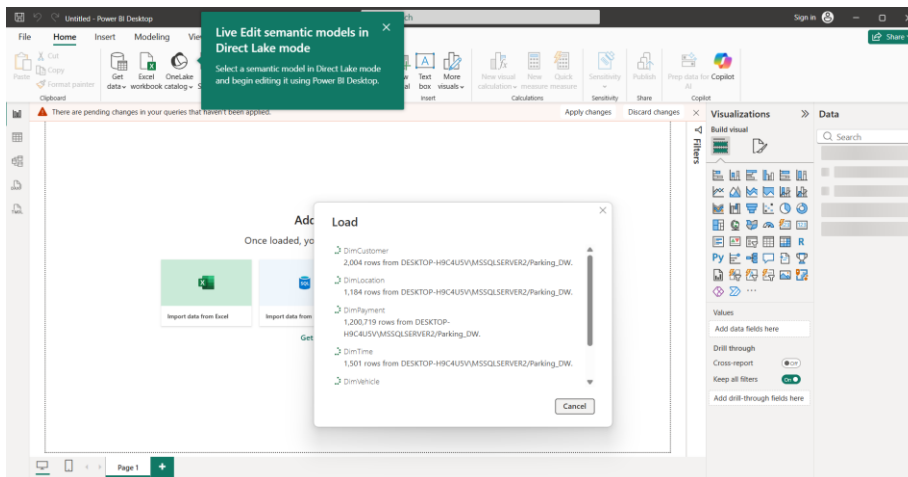
- Entered the Analysis Services server name and selected the ParkingCube database.



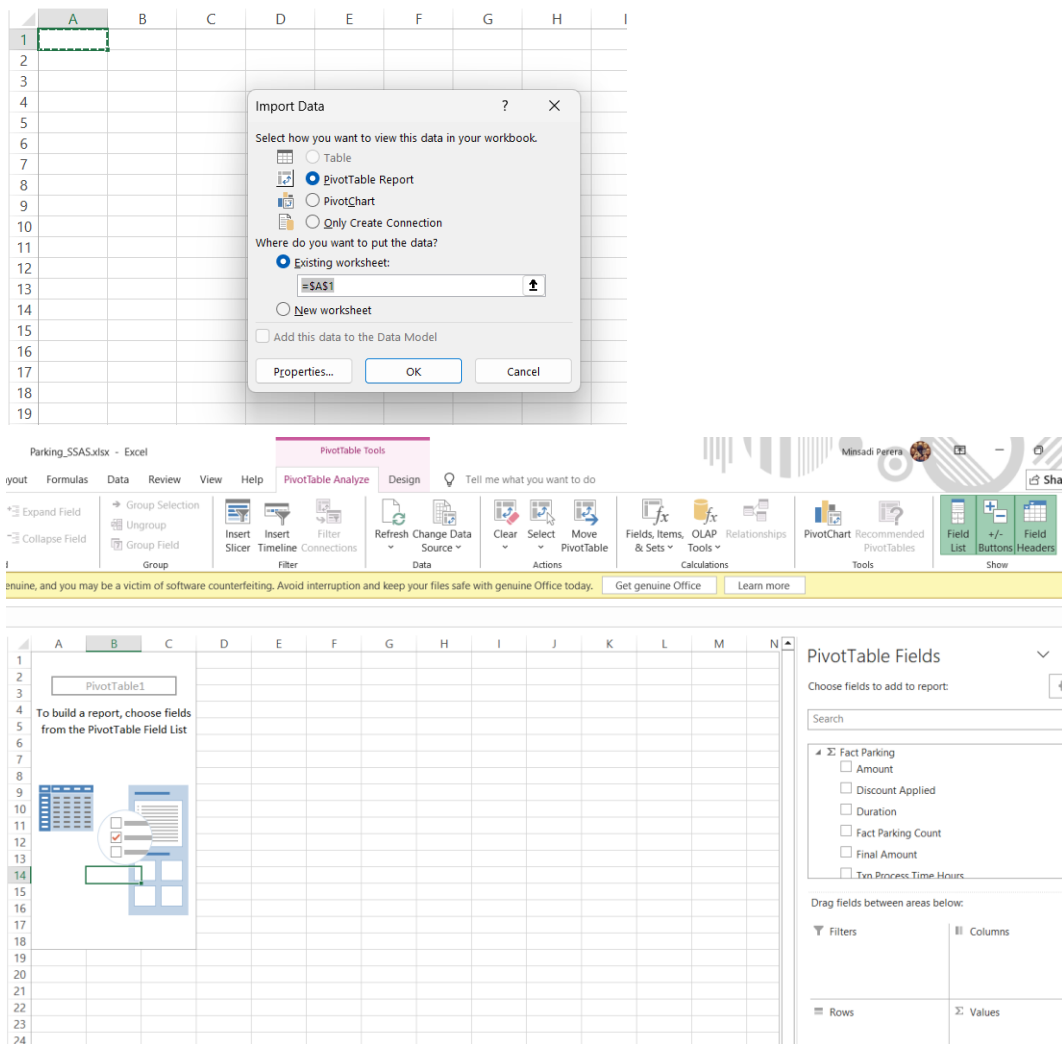
- Selected the cube and imported the connection into the workbook.



- Load all data in dimensions tables and fact tables



- Inserted a PivotTable using the SSAS cube as the data source.



3.2 Roll-Up

Roll-up is an OLAP operation used to aggregate data to a higher level in a hierarchy, reducing detail and showing summarized information.

In this Pivot Table, roll-up is demonstrated by aggregating detailed parking transaction data into higher-level summaries.

Individual transactions are grouped by payment method and further summarized by year. The Grand Total represents the highest level of aggregation, combining all years and payment methods into a single total.

This shows how data is rolled up from detailed records to yearly and overall summaries.

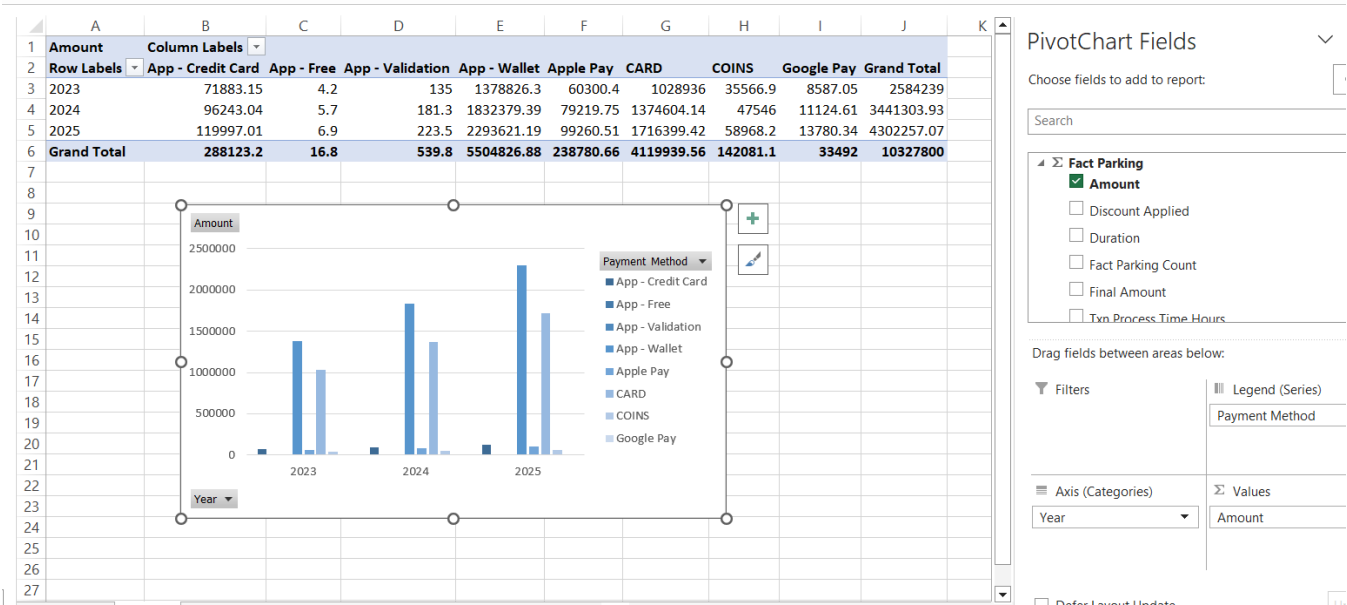


Figure : Roll-Up – Monthly parking revenue aggregated to yearly totals

3.3 Drill-Down

Drill-down is used to move from summarized data to more detailed levels within a hierarchy.

In this Pivot Table, drill-down is demonstrated by expanding summarized yearly data into more detailed levels such as months and individual dates.

The data initially app

ears at the year level (2023, 2024, 2025), but by expanding it, you can view detailed transaction amounts for specific days (e.g., 26, 27, 28) under each year.

This shows how drill-down allows users to move from high-level summaries to more granular data, enabling deeper analysis of individual transactions within the time hierarchy.

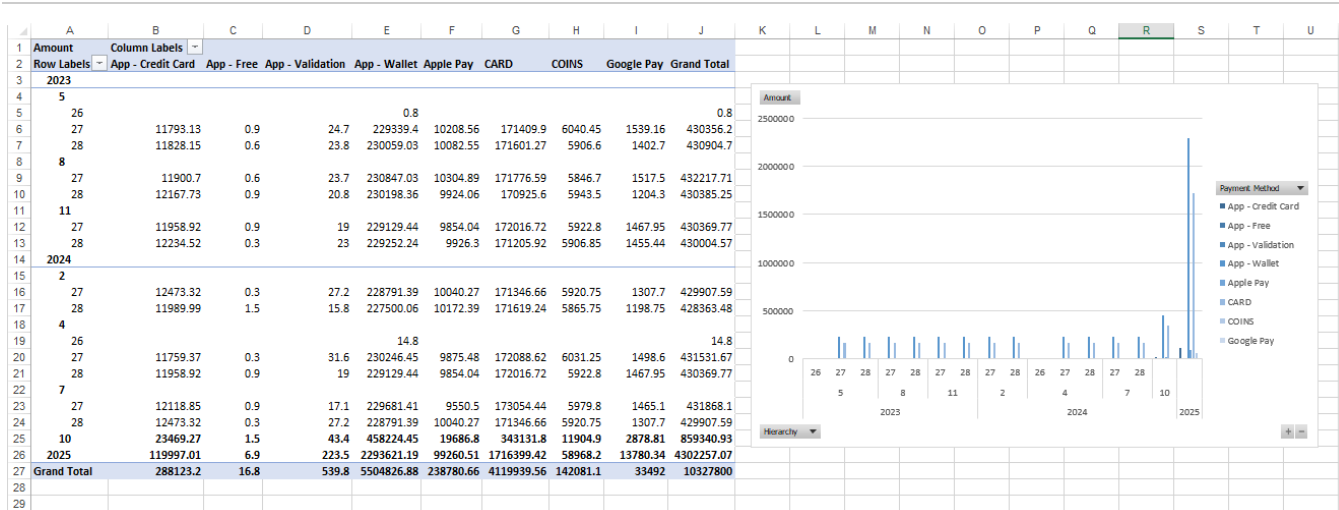


Figure : Drill-Down – Yearly revenue expanded to monthly detail

3.4 Slice

Slice is an operation that filters data based on a single dimension.

In this Pivot Table, slice is demonstrated by filtering the data using a specific value of a single dimension, which is the Payment Method.

Here, the data is filtered to show only “App - Credit Card”, so all the values displayed (amounts by location group) belong only to that selected payment method.

This allows users to focus on a particular subset of data without changing the overall structure of the Pivot Table, making it easier to analyze how one payment method performs across different locations

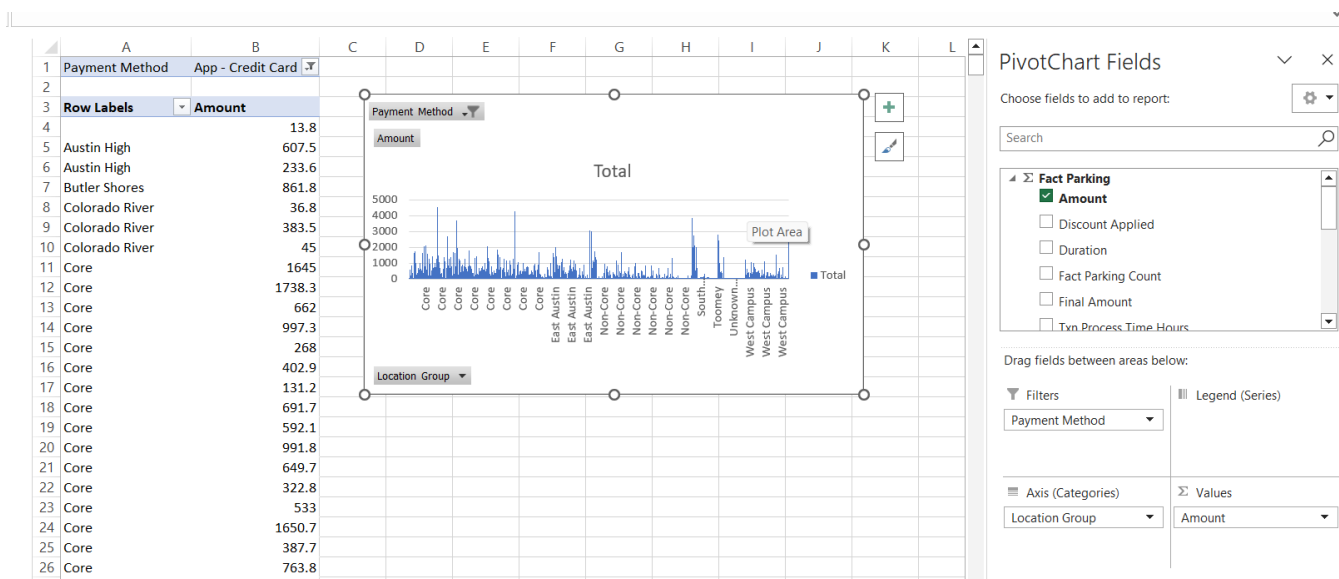


Figure : Slice – Filtering by Payment Method = Credit Card

3.5 Dice

The Dice operation selects values from two or more dimensions simultaneously.

In this Pivot Table, dice is demonstrated by filtering the data using multiple dimensions simultaneously, specifically Year (2023) and Payment Method (App – Credit Card).

This creates a smaller, focused subset of the data where only transactions that match both conditions are displayed.

The table then analyzes this subset further by showing the total amount for each vehicle type (e.g., Convertible, Coupe, SUV).

This illustrates how dicing allows users to examine data from multiple perspectives at once for more specific and targeted analysis.

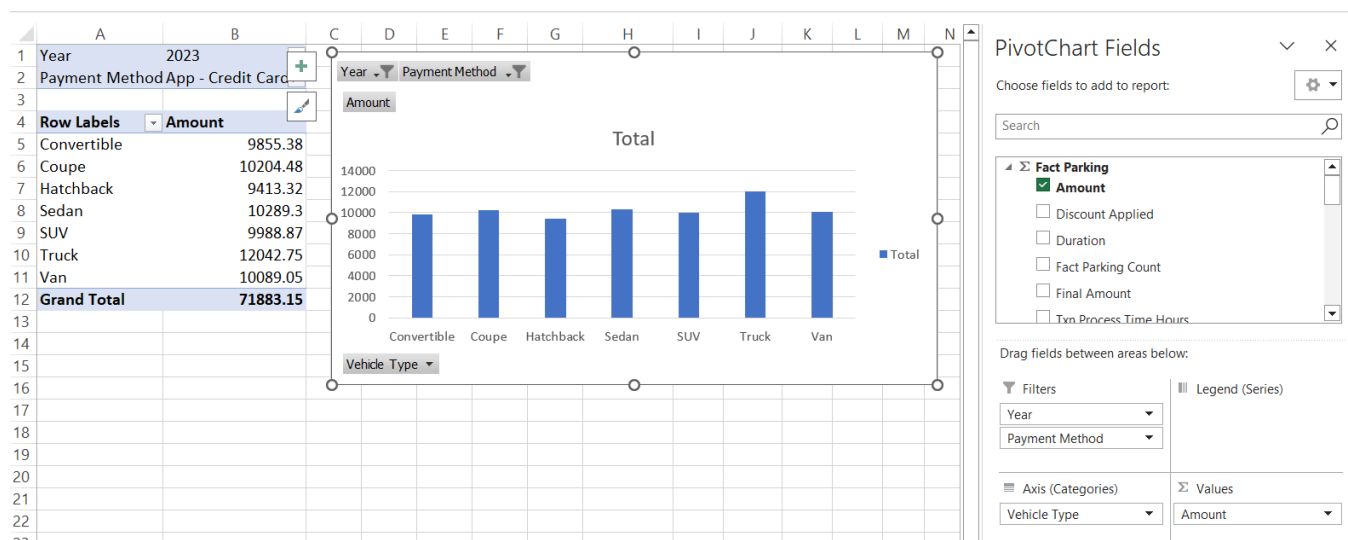


Figure : Dice – Filtering by Year range and Location Group

3.6 Pivot

Pivot is an operation that reorganizes data by rotating rows and columns to view it from different perspectives.

In this PivotTable, pivoting is demonstrated by swapping dimensions such as Vehicle Type and Location Group between rows and columns to analyze the same data in different layouts.

Here, Vehicle Type is placed in the columns (legend), while Location Group is in the rows (axis), and the Amount is summarized for each combination. This rearrangement allows users to compare how different vehicle types perform across various locations.

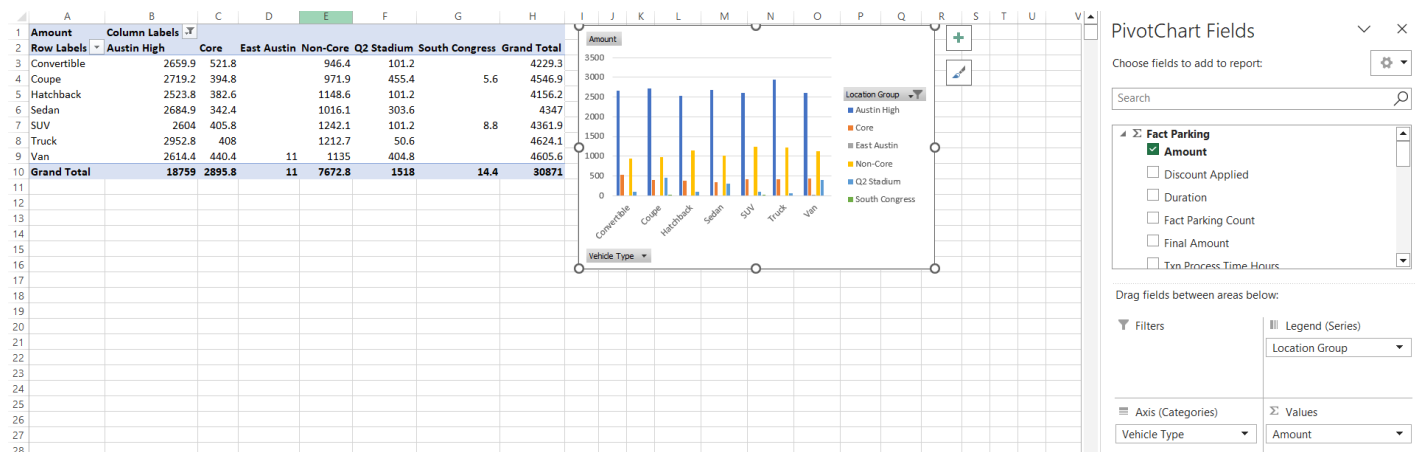
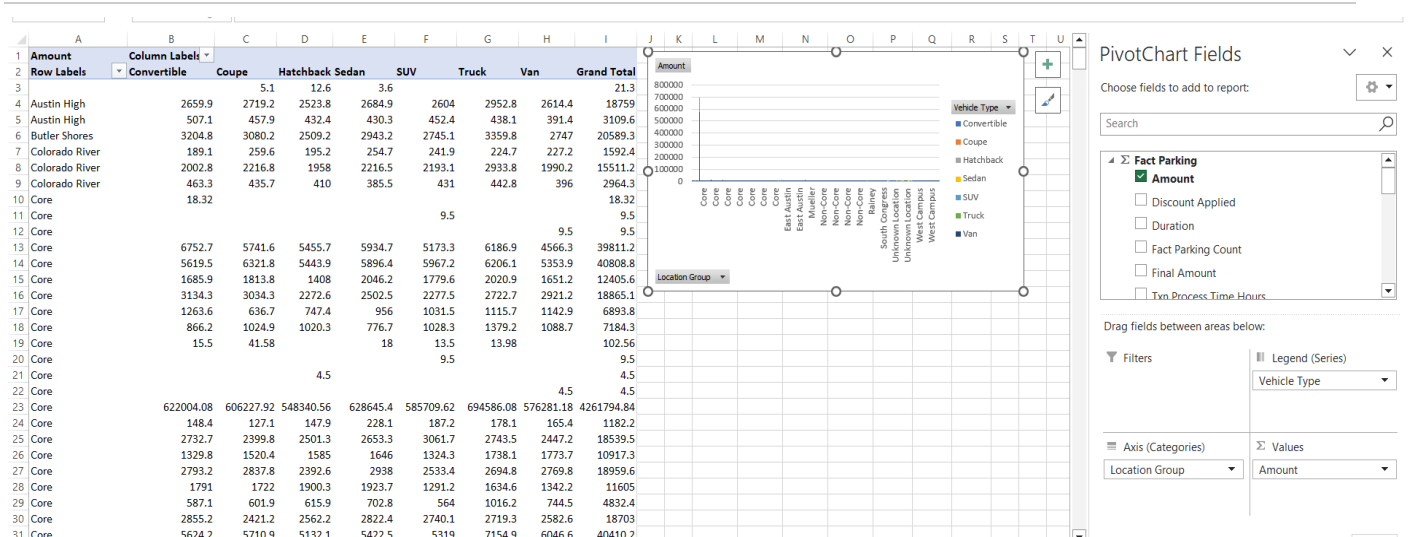


Figure : Pivot – Rotating dimensions for alternative view

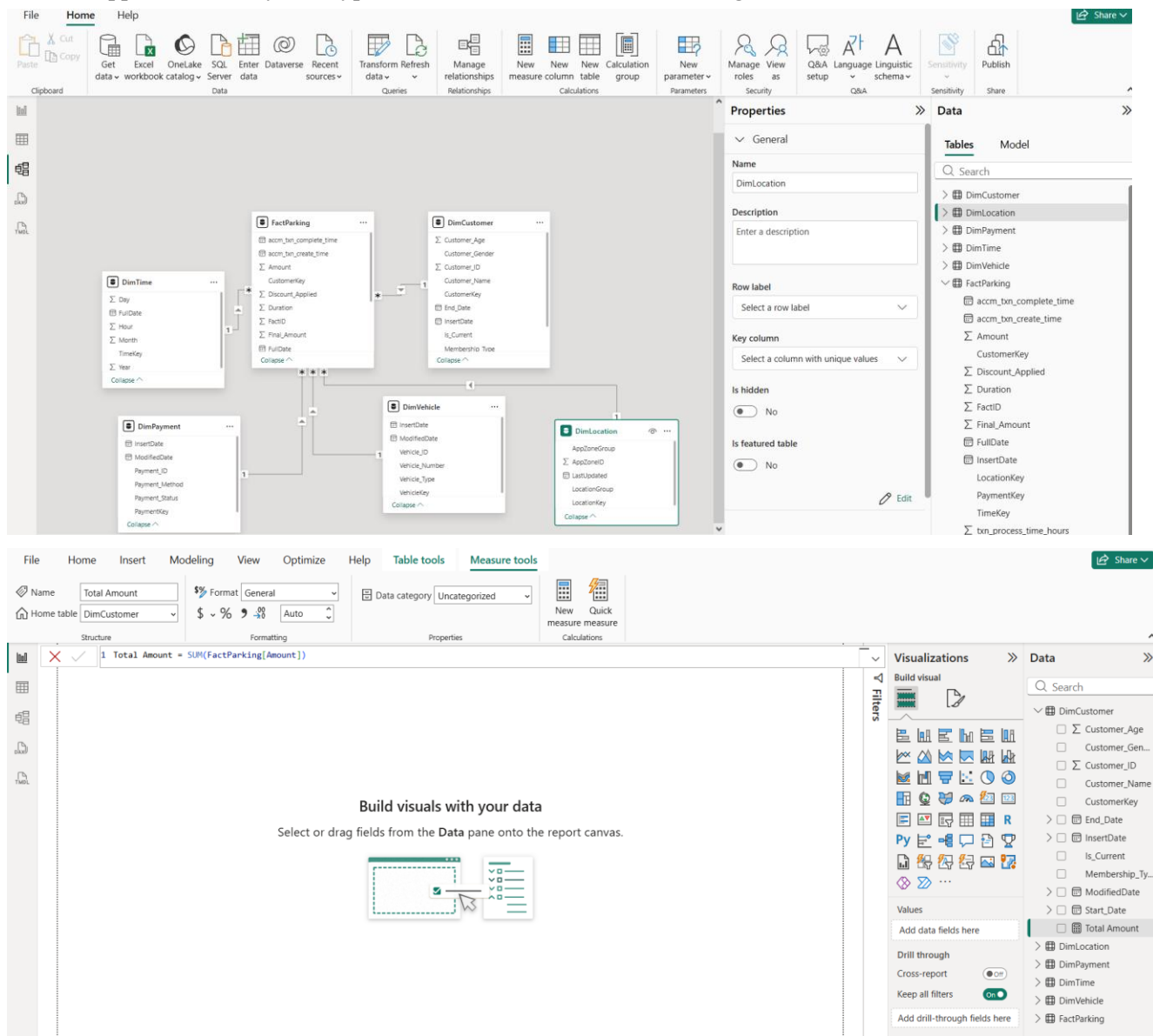
Step 4: Power BI Reports

Four Power BI reports were developed and published to Power BI Service. Each report demonstrates different data visualization and analytical capabilities using the Parking_DW data warehouse as the data source.

4.1 Connecting Power BI to the Data Warehouse

Power BI Desktop was connected to the Parking_DW SQL Server database. The following steps were performed:

- Opened Power BI Desktop and selected "Get Data" → "SQL Server".
- Entered the SQL Server instance name and selected Parking_DW.
- Imported the FactParking, DimCustomer, DimVehicle, DimLocation, DimPayment, and DimTime tables.
- Verified relationships between tables in the Model view.
- Applied necessary data type conversions and column formatting.



4.2 Report 1: Matrix Visual with Row and Column Groupings

This report uses a Matrix visual to present detailed tabular data with both row and column groupings.

The matrix provides a cross-tabular view of total parking transaction amounts by Year (rows) and Payment Method (columns), allowing users to easily compare revenue across different payment methods over time.

The visual also supports drill-down within the Year hierarchy, enabling analysis at more detailed levels such as months.

Data Preparation

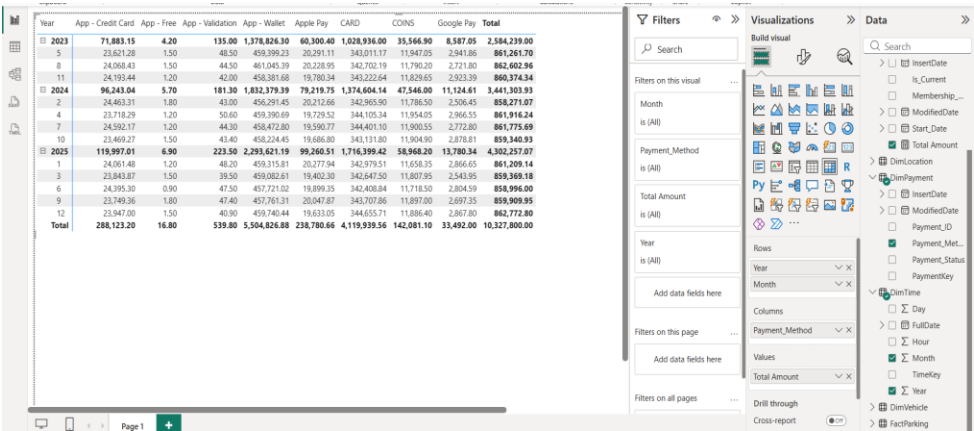
The FactParking table was used along with relevant dimensions. A DAX measure was created to calculate total revenue:

$$\text{Total Amount} = \text{SUM}(\text{FactParking}[\text{Amount}])$$

Visual Design

The Matrix visual was configured as follows:

- **Rows:** Year (with drill-down to months)
- **Columns:** Payment Method
- **Values:** Total Amount(Amount)



Parking Transactions Matrix By Payment Method and Year

10.33M

Total Amount

Total transactions Payment amount by each Payment_Method

Year	App - Credit Card	App - Free	App - Validation	App - Wallet	Apple Pay	CARD	COINS	Google Pay	Total
2023	71,883.15	4.20	135.00	1,378,826.30	60,300.40	1,028,936.00	35,566.90	8,587.05	2,584,239.00
2024	96,243.04	5.70	181.30	1,832,379.39	79,219.75	1,374,604.14	47,546.00	11,124.61	3,441,303.93
2025	119,997.01	6.90	223.50	2,293,621.19	99,260.51	1,716,399.42	58,968.20	13,780.34	4,302,257.07
2026	146,061.48	1.20	48.20	459,315.81	20,277.94	342,979.51	11,658.35	2,866.65	861,209.14
2027	23,843.87	1.50	39.50	459,082.61	19,402.30	342,647.50	11,807.95	2,543.95	859,369.18
2028	24,395.30	0.90	47.50	457,721.02	19,899.35	342,408.84	11,718.50	2,804.59	858,996.00
2029	23,749.36	1.80	47.40	457,761.31	20,047.87	343,707.86	11,897.00	2,697.35	859,909.95
2030	23,947.00	1.50	40.90	459,740.44	19,633.05	344,655.71	11,886.40	2,867.80	862,772.80
Total	288,123.20	16.80	539.80	5,504,826.88	238,780.66	4,119,939.56	142,081.10	33,492.00	10,327,800.00

Total Amount by Payment_Method



4.3 Report 2: Multiple Slicers with Cascading Filters

This report demonstrates the use of multiple slicers with cascading filter functionality to analyze parking transaction data interactively. Users can filter the data using slicers such as Location Group and Vehicle Type, where selections dynamically update all visuals in the report. The cascading effect ensures that selecting values in one slicer limits the relevant options and data shown in other slicers and visuals.

Data Preparation

The report uses the FactParking table along with related dimension tables such as DimLocation and DimVehicle. Relationships between these tables enable proper filtering and interaction across visuals.

Cascading Filter Implementation

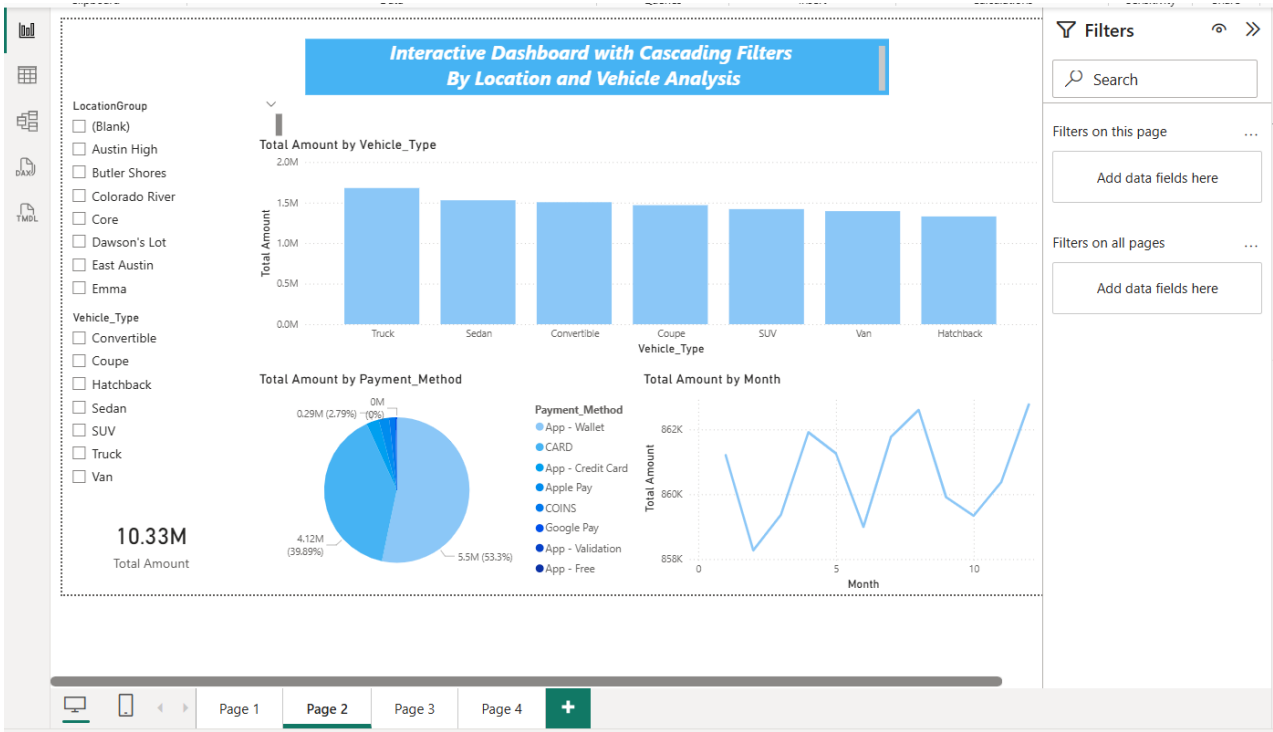
Cascading filters were implemented as follows:

- Slicer 1 (Location Group): Displays available locations (e.g., Austin High, Core, Downtown).
- Slicer 2 (Vehicle Type): Displays vehicle categories (e.g., Truck, Sedan, SUV).
- Selecting a value in one slicer filters the data and updates the available context in other slicers and visuals.
- Power BI's default filtering behavior enables this cascading interaction across visuals.

Visual Design

The report includes:

- **Bar Chart:** Total Amount by Vehicle Type
- **Pie Chart:** Total Amount by Payment Method
- **Line Chart:** Total Amount by Month (trend analysis)
- **Card Visual:** Total Amount (10.33M)



4.4 Report 3: Drill-Down Report with Time Hierarchy

This report demonstrates drill-down functionality using a time hierarchy, allowing users to explore parking transaction data from a summarized level to more detailed levels. Users can navigate through Year → Month → Day to analyze how total transaction amounts change over time.

Data Preparation

The DimTime table was used to create the hierarchy with Year, Month, and Day fields. The FactParking table provides the measure:

Total Amount = SUM(FactParking[Amount])

Drill-Down Implementation

Drill-down was implemented using a column chart displaying total amount across the time hierarchy. The hierarchy levels are:

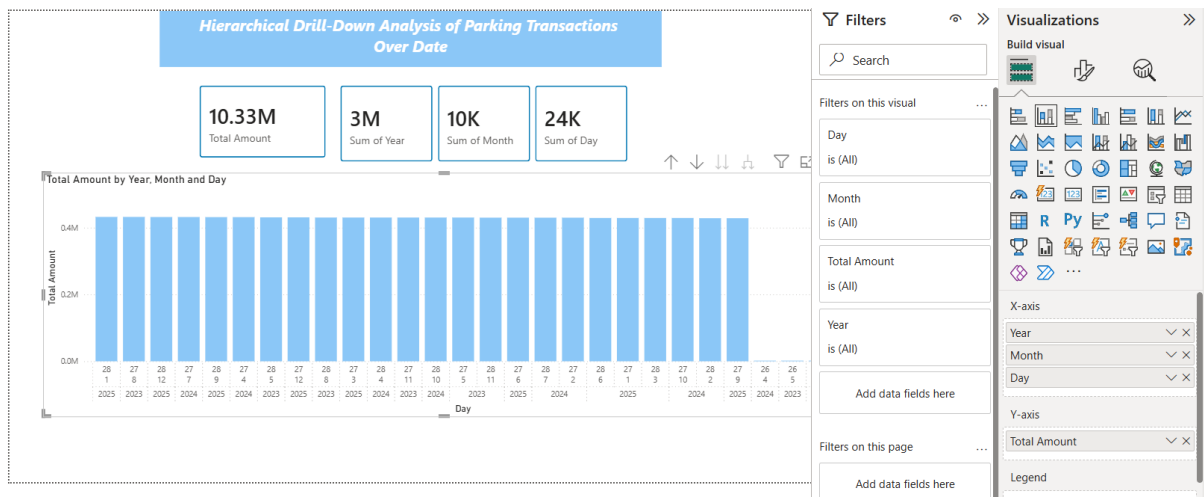
- Level 1: Year – shows total amount aggregated annually
- Level 2: Month – shows totals within each selected year
- Level 3: Day – shows detailed daily transaction amounts

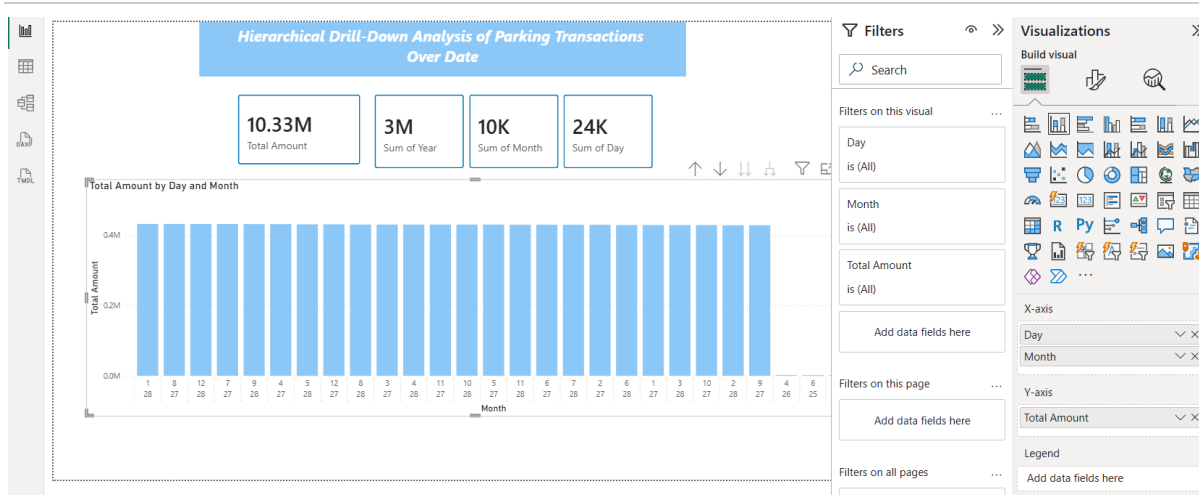
Users can use the drill-down and expand buttons on the chart to move between these levels and explore data in detail.

Visual Design

The report includes:

- **Column Chart:** Total Amount by Year, Month, and Day
- **Card Visuals:**
 - Total Amount (10.33M)
 - Sum of Year (3M)
 - Sum of Month (10K)
 - Sum of Day (24K)





4.5 Report 4: Drill-Through Report

➤ Summary Report (Drill-Through Source Page)

The summary report provides a **high-level overview** of parking transaction data using multiple visuals. It allows users to quickly analyze key patterns and then navigate to a more detailed view using drill-through functionality.

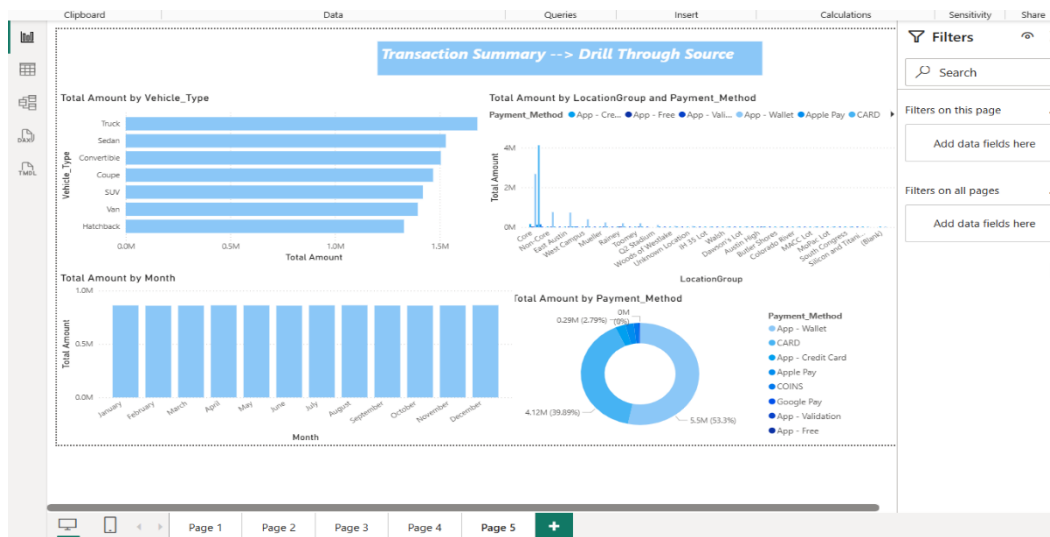
Visual Design

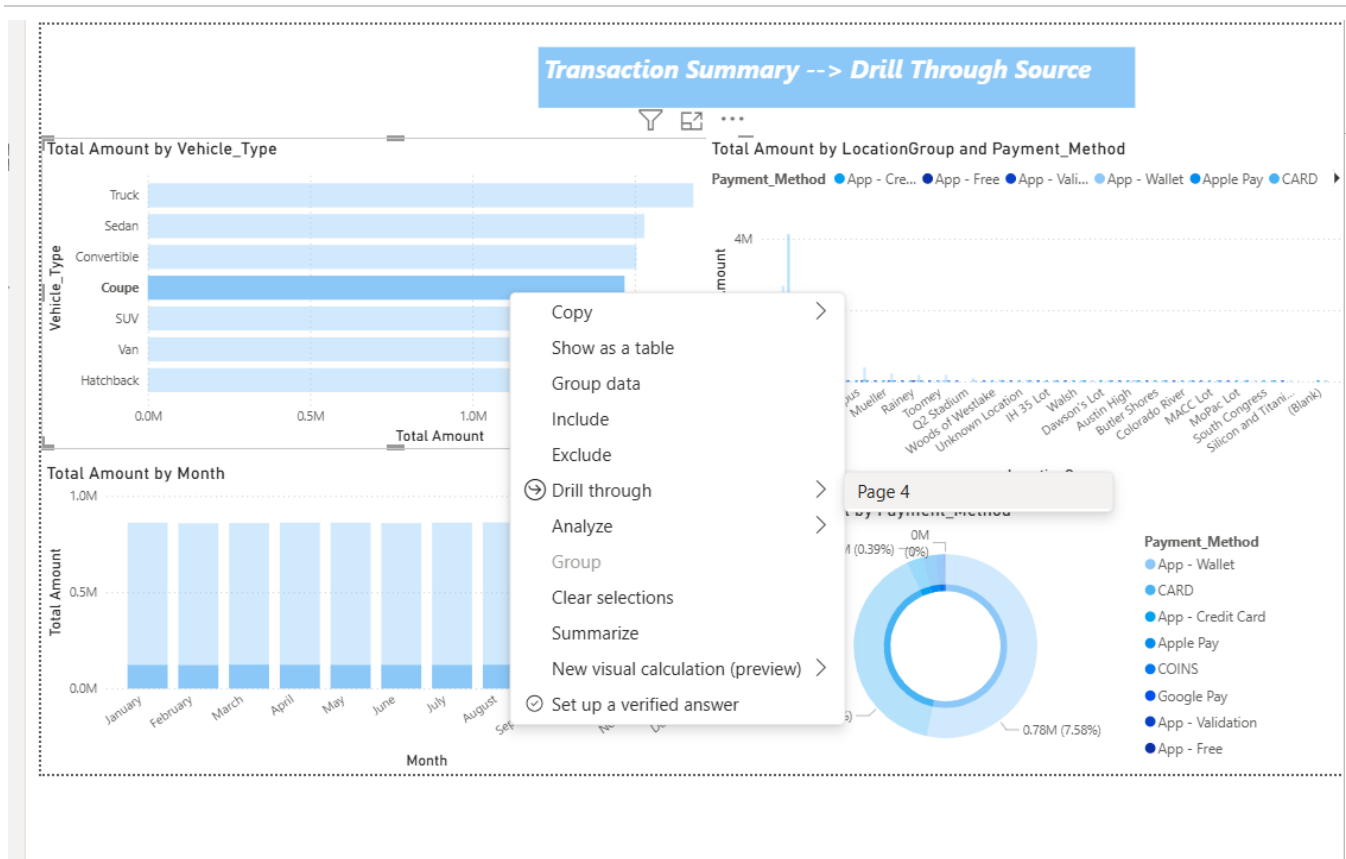
The summary page includes:

- **Bar Chart:** Total Amount by Vehicle Type
- **Column Chart:** Total Amount by Location Group and Payment Method
- **Line Chart:** Total Amount by Month (trend analysis)
- **Donut Chart:** Total Amount by Payment Method

Functionality

- Users can **right-click on any data point** (e.g., a vehicle type such as Convertible).
- Select **“Drill Through”** → **Detail Page**.
- This action passes the selected value as a filter to the detailed report.





➤ Detailed Report (Drill-Through Target Page)

The detailed report displays **transaction-level data** filtered based on the selection made in the summary report. It provides a deeper view of the selected category.

Drill-Through Configuration

- The field **Vehicle Type** is added to the **Drill-through filters** pane.
- When a user drills through, the page is automatically filtered for the selected vehicle type (e.g., Convertible).

Visual Design

The detail page includes:

- **Table Visual:**
Displays detailed fields such as:
 - Vehicle Type
 - Location Group
 - Payment Method
 - Total Amount
 - Year, Quarter, Month, Day
- **Bar Chart:** Total Amount by Vehicle Type (filtered view)

- **Back Button:** Allows navigation back to the summary page

The screenshot shows a Power BI report interface. On the left, a data table is displayed with columns: Vehicle_Type, LocationGroup, Payment_Method, Total Amount, Year, Quarter, Month, and Day. The table contains 27 rows of data for various vehicle types and locations. On the right, there is a sidebar with 'Filters' and 'Visualizations' sections. The 'Filters' section includes a search bar and a list of filters for 'FullDate - Day', 'FullDate - Month', 'FullDate - Quarter', 'FullDate - Year', 'LocationGroup', 'Payment_Method', and 'Total Amount'. The 'Visualizations' section includes a 'Build visual' button and a list of visual types like 'Table', 'Card', 'Bar chart', etc. The 'Data' section on the far right shows a list of data fields with checkboxes for selection.

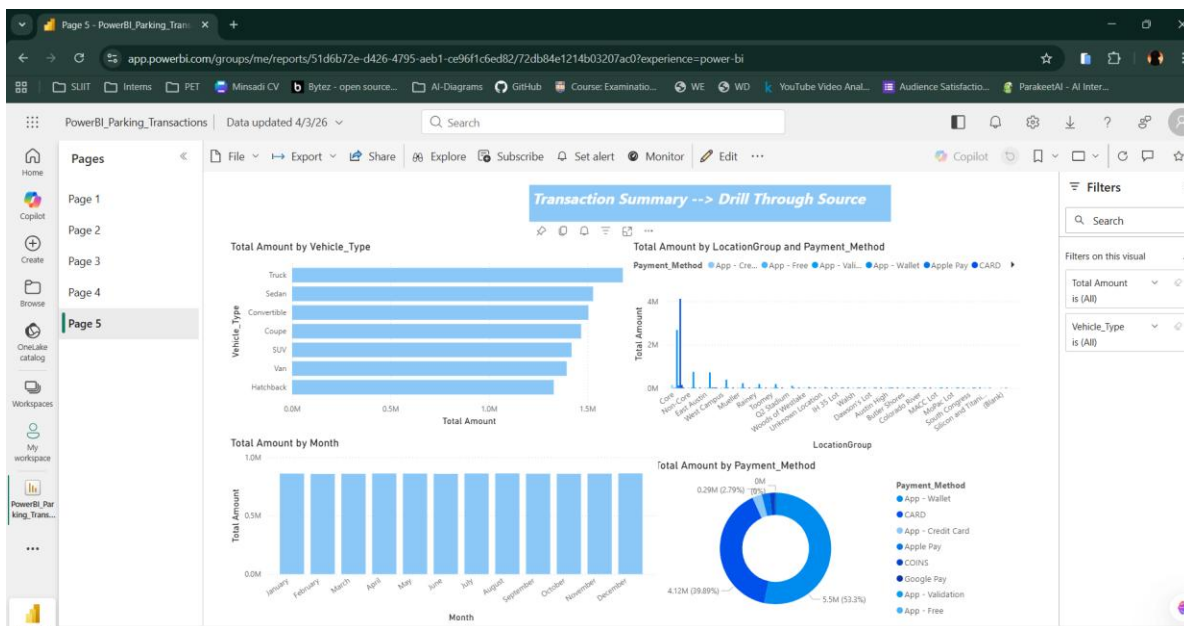
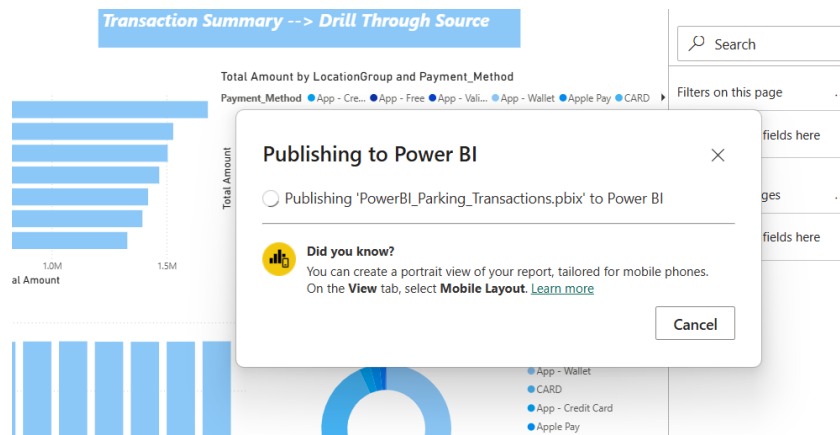
The screenshot shows a Power BI report titled 'Transaction Detail - Drill- Through Report'. It features a 'Back To Summary' button at the top left. Below the button is a data table with columns: Vehicle_Type, LocationGroup, Payment_Method, Total Amount, Year, Quarter, Month, and Day. The table contains 27 rows of data for various vehicle types and locations. To the right of the table is a bar chart titled 'Total Amount by Vehicle_Type'. The chart shows a single bar for 'Convertible' with a value of approximately 1.5M. The x-axis is labeled 'Total Amount' and ranges from 0.0M to 2.0M. The y-axis is labeled 'Vehicle_Type' and has a single category 'Convertible'. At the bottom, there is a navigation bar with buttons for 'Page 1', 'Page 2', 'Page 3', 'Page 4' (selected), and 'Page 5'.

The screenshot shows a Power BI report titled 'Transaction Detail - Drill- Through Report'. It features a 'Back To Summary' button at the top left. Below the button is a data table with columns: Vehicle_Type, LocationGroup, Payment_Method, Total Amount, Year, Quarter, Month, and Day. The table contains 27 rows of data for various vehicle types and locations. To the right of the table is a bar chart titled 'Total Amount by Vehicle_Type'. The chart shows a single bar for 'Convertible' with a value of approximately 1.5M. The x-axis is labeled 'Total Amount' and ranges from 0.0M to 1.5M. The y-axis is labeled 'Vehicle_Type' and has a single category 'Convertible'. At the bottom, there is a navigation bar with buttons for 'Page 1', 'Page 2', 'Page 3', 'Page 4' (selected), and 'Page 5'.

4.6 Publishing Reports to Power BI Service

All four reports were published to Power BI Service using the following steps:

- Saved the Power BI Desktop file (.pbix).
- Clicked "Publish" in the Home ribbon.
- Selected the target workspace in Power BI Service.
- Opened Power BI Service in a browser and confirmed the reports were successfully published.
- Each report was verified by opening it in the browser and confirming all visuals loaded correctly.



Summary

This report documents the implementation of Assignment 2 for the Data Warehousing and Business Intelligence (IT3021) module. The following tasks were completed:

- Step 1: The Parking_DW data warehouse from Assignment 1 was used as the data source, with its star schema comprising FactParking and five dimension tables.
- Step 2: An SSAS Multidimensional cube was created with measures, dimensions, and hierarchies implemented for the Location and Time dimensions.
- Step 3: Excel was connected to the SSAS cube and all five OLAP operations (Roll-up, Drill-down, Slice, Dice, Pivot) were demonstrated using PivotTables.
- Step 4: Four Power BI reports were developed (Matrix, Cascading Slicers, Drill-Down, Drill-Through) and published to Power BI Service.